

YOONJI CHOI

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[LinkedIn](#) • [GitHub](#) • [Website](#)

EDUCATION

New York University Courant Institute
Master of Science in Computer Science

New York, New York
Expected May 2027

Stony Brook University

Bachelor of Science in Applied Mathematics and Statistics
Minor in Computer Science | Honors: Cum Laude, Dean's List, Merit Scholarship

Stony Brook, New York
February 2022 – May 2025
GPA: 3.63

SKILLS

Programming Languages: Python, Java, R, SAS, MATLAB

Core CS: Data Structures, Algorithms, Object-Oriented Programming, Parallel Computing (OpenMP, MPI)

Tools: scikit-learn, Streamlit, NLP (TF-IDF, SHAP, SMOTE)

License: Advanced Data Analytics Semi-Professional (Korea Data Agency, 2024)

PROJECTS

Minimum vs. Living Wage by U.S. State

April 2025 - May 2025

- Analyzed wage and socioeconomic data for 50 U.S. states, applying EDA and statistical tests (Linear regression, t-test) to assess wage gap significance.
- Performed K-means clustering on living wage and SNAP participation features, identifying distinct state groups with high poverty vulnerability and visualizing disparities.

Olist Return Analysis

February 2025 - May 2025

- Built a SMOTE-enhanced Random Forest model on 100K+ e-commerce orders from the Olist Kaggle dataset, achieving 95% accuracy, and developed a Streamlit dashboard with SHAP integration for interpretability.
- Processed 50K+ Portuguese customer reviews using Google Translate API and extracted TF-IDF features for predictive modeling.

Energy Usage & Weather Data Analysis

March 2024 – April 2024

- Preprocessed and merged large-scale energy consumption and weather datasets, applying time-series transformations and daily aggregation in Python.
- Engineered energy and weather features to explore correlations and trends, reinforcing skills in data cleaning, pipeline design, and exploratory analysis.

EXPERIENCE

Stony Brook University | Undergraduate Research Assistant

January 2025 – May 2025

- Implemented and applied HPC techniques (OpenMP, MPI) to PDE/ODE solvers relevant to turbulent mixing simulations, achieving up to 1.6× faster runtime in large-scale numerical computations.
- Analyzed performance scalability of serial vs. parallel solvers, ensuring accuracy in finite difference models and contributing to the verification process of turbulent mixing methods.

Stony Brook University | Undergraduate Teaching Assistant & Tutor

August 2023 - June 2024

- Facilitated learning for ~100 students in Calculus, Linear Algebra, and Statistics, reinforcing mathematical reasoning and analytical problem-solving skills.
- Provided personalized tutoring while collaborating with faculty on exams, grading, progress tracking, and records management, developing strong organizational and communication skills.

EXTRACURRICULAR ACTIVITIES

FORC Summer Camp | Configuring Generative AI for Research Using Python

August 2025

- Trained in Generative AI and Retrieval-Augmented Generation (RAG) for research applications, building Python-based pipelines with embedding, retrieval, and prompt augmentation.
- Applied Portkey API to configure and deploy LLM workflows, gaining hands-on experience with API integration and responsible AI practices.